

Advanced Aeronautics Report

Summary

This coursework aims to compare the aerodynamic performance of a generated Karman-Trefftz(KT) airfoil to the geometrically most similar NACA airfoil, using coding and modelling to compare the 2 airfoils. The KT airfoil is generated by setting parameters for its thickness, camber, and trailing edge angle, and the change in geometry of the airfoil is investigated as each of these parameters vary. Then the doublet panel method is used to estimate the lift produced by the airfoil, and the accuracy of the method is evaluated through a comparison of theoretical values and numerical predictions. Finally, the X-Foil software is used to compare the aerodynamic performance of the 2 airfoils, measuring different quantities that provide information about the designs of the airfoils. The parameters are changed such as surface roughness and the ratio of inertial to viscous forces of the flow, allowing for a more thorough comparison.

Q1 - Karman-Trefftz Airfoil Generation and Analysis

KT airfoils are generated by applying conformal mapping to transform circles onto a complex plane by controlling three main geometric properties: ε (thickness), β (camber), and τ (trailing-edge angle). For the assigned values of $\varepsilon = 0.07$, $\beta = 0.07$, $\tau = 0.05$, the generated airfoil had moderate thickness and mild camber. Using the extracted co-ordinate distribution, the airfoil's maximum thickness and camber, along with their chordwise locations, are calculated directly from the geometry. This allows for the maximum thickness position and camber of the airfoil to be observed as seen in figure 1 and compared to a NACA airfoil.

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=== Q1(a): KT geometry ===  
Max camber      = 0.0348 at x/c = 0.502  
Max thickness   = 0.0946 at x/c = 0.278  
Closest NACA 4-digit ≈ NACA 3509
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Fig. 1: KT Airfoil Geometry Values

The key values of geometry for the KT airfoil as seen in figure 1 were then converted to the nearest 10% for thickness x/c , and nearest 1% for camber x/c and actual thickness value. This results in the NACA 3509 being the closest geometric airfoil to the generated KT airfoils. Figure 2 shows how they have similar global thickness trends, but there are key differences in the profile at the leading and trailing edges due to the mapping characteristics of the Karman-Trefftz method.

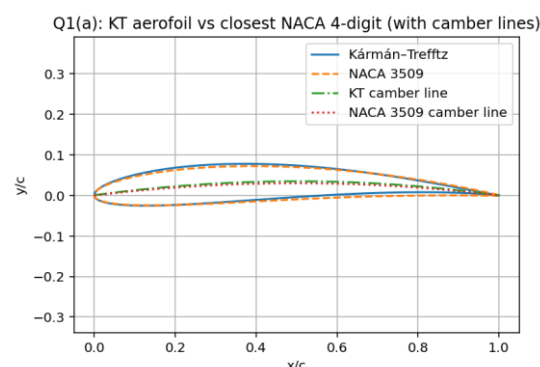


Fig. 2 KT Airfoil and Naca Airfoil Overlaid

The thickness and camber distributions along the chord provide further insight into how the parameters influence the KT airfoil. The thickness distribution is primarily governed by ϵ , showing a smooth increase to a single maximum before tapering to the trailing edge. The camber curve reflects the influence of β , with curvature increasing as β increases. These distributions are essential for matching KT shapes to equivalent NACA airfoils and for understanding how each parameter locally affects the profile.

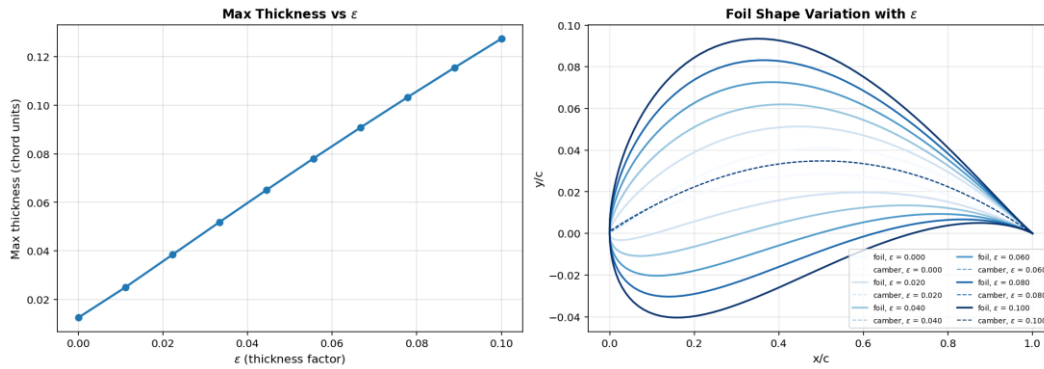


Fig.3 Plots of KT Airfoil with varying ϵ

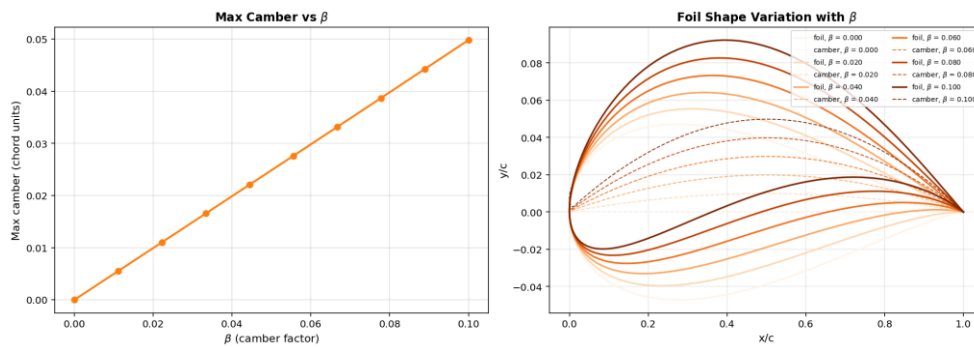


Fig.4 Plots of KT Airfoils with varying β

Figure 3 demonstrates that ϵ acts almost exclusively as a thickness-scaling parameter in the Kármán–Trefftz formulation. The maximum thickness increases linearly with ϵ , confirming a direct and proportional influence on the vertical separation between the upper and lower surfaces after mapping. The corresponding airfoil shapes show that thickness grows while the camber profile remains essentially unchanged, indicating that ϵ does not affect the mean-line curvature. Figure 4 shows that β primarily governs camber. The maximum camber increases linearly with β , reflecting the increasing asymmetry introduced by the angular offset in the circle-plane mapping. The generated airfoils exhibit progressively higher camber while maintaining nearly constant thickness, confirming that β influences curvature distribution rather than overall cross-sectional size. The shapes remain smooth and well-behaved across the full parameter range, demonstrating that the KT mapping provides controlled and independent adjustment of camber without distorting other geometric features.

Q2 - Doublet Panel Aerodynamic Performance Analysis

The doublet panel method is a boundary-element technique used to model incompressible, irrotational flow around aerodynamic bodies. The airfoil surface is divided into a series of straight-line panels, each carrying a constant-strength doublet. The velocity potential induced by all panels must satisfy the flow tangency condition, which requires the normal component of velocity at every control point on the surface to be zero. A wake panel is added at the trailing edge to enforce the Kutta condition, ensuring that flow leaves the airfoil smoothly. Solving the resulting linear system yields the doublet strengths, from which the surface pressure coefficient distribution C_p and the overall lift coefficient C_L are obtained.

To assess the order of accuracy, the doublet panel method was applied to the KT airfoil defined by $\varepsilon = 0.07$, $\beta = 0.07$, and $\tau = 0.05$ at an incidence angle of $\alpha = 3^\circ$. The method was evaluated for increasing numbers of surface panels, from 20 to 320. For each case, the computed lift coefficient was compared to the theoretical value obtained from the KT mapping. As the number of panels increases, the discretisation better resolves the curvature of the airfoil surface and the pressure gradient near the leading and trailing edges, resulting in a more accurate estimate of C_L .

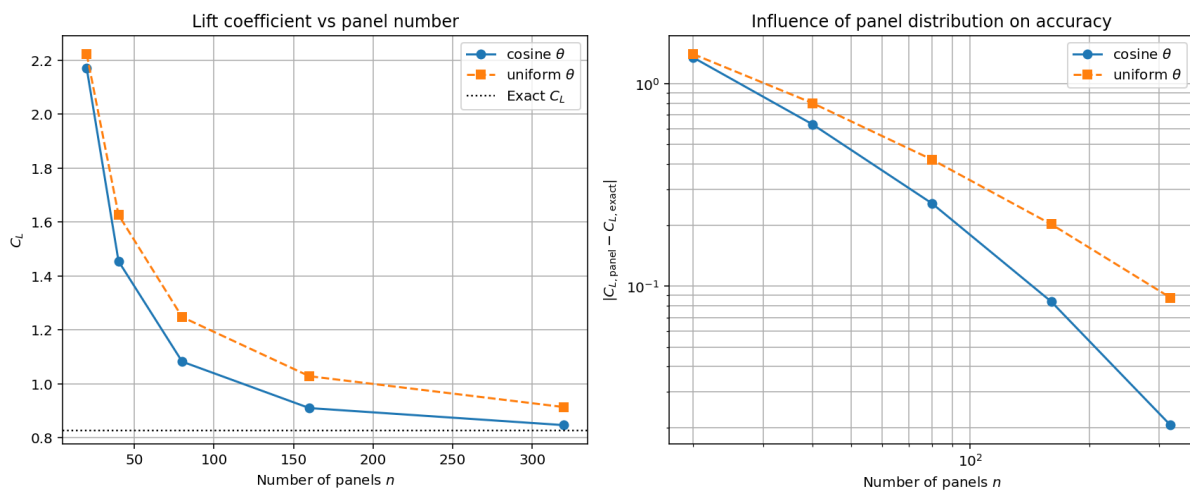


Fig.5 Plots of C_L and C_L Error vs Panel Number for both methods

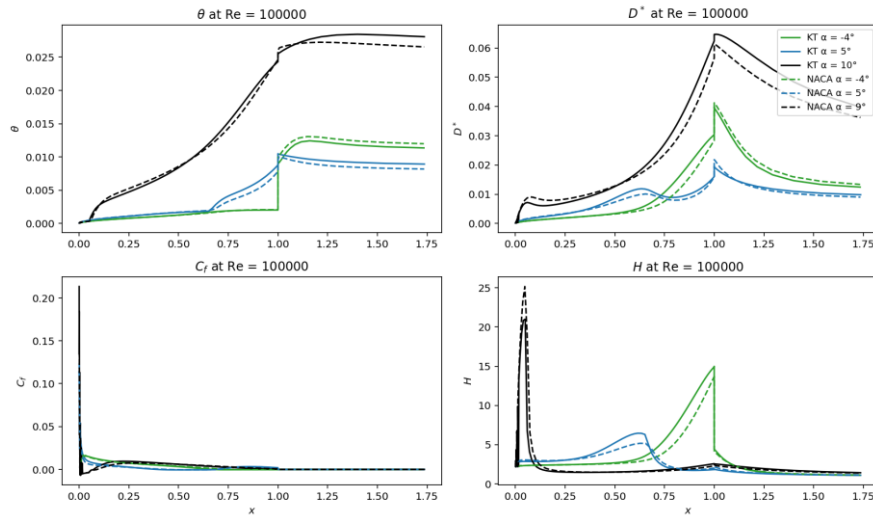
The results show that the doublet panel method converges toward the theoretical lift coefficient obtained from conformal mapping as the number of panels is increased. For both panel distributions, C_L initially deviates significantly from the exact value at low panel counts due to poor geometric resolution of the sharp leading-edge curvature. As the number of panels increases, the predicted C_L approaches the exact value asymptotically, demonstrating the expected logarithmic convergence behaviour of low-order panel methods. A log-log fit of error versus panel size confirms an order of accuracy of approximately $\mathcal{O}(h^p)$, consistent with the underlying piecewise-constant doublet approximation scheme.

Comparing the cosine- θ and uniform- θ distributions, the influence of panel clustering is clearly visible. The cosine distribution consistently yields lower error at every panel count because it places a higher density of panels near the leading and trailing edges, where geometric curvature and pressure gradients are highest. This leads to a more accurate representation of the

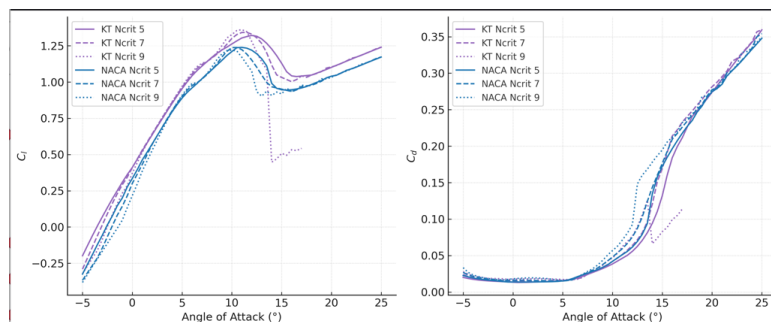
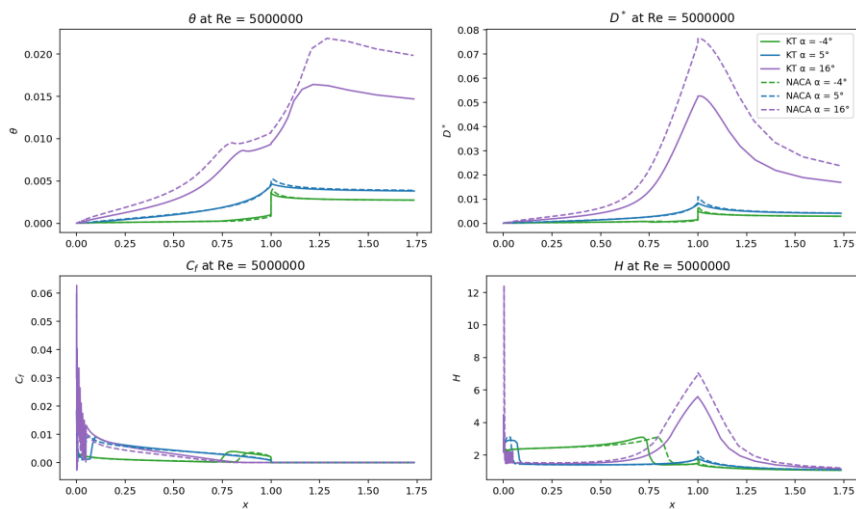
Kármán–Trefftz airfoil geometry and its local flow features. In contrast, the uniform distribution resolves the surface poorly in these regions, resulting in larger errors and slower convergence. Overall, the analysis shows that panel distribution has a significant impact on accuracy, and cosine spacing is strongly preferable for smooth aerodynamic surfaces such as KT airfoils.

Q3&4 - X-Foil Analysis

Aerodynamic parameters at $Re = 1e+05$



Aerodynamic parameters at $Re = 5e+06$



Conclusion